PionCloud v1.4 — changes and improvements

PionCloud maintainers (TDIS)

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branch `v1.4-improvements` of https://github.com/ayerbeg/PionCloud

This document lists *every* change made in v1.4 with respect to v1.3 (commit `3254880` plus the manual update merged into `main`). For each change it states **what** changed, **why**, the **evidence**, whether **numbers change** and by how much, and how to **get the old behaviour**. The same information, in short form, is in `CHANGELOG.md`; the user manual (`docs/PionCloud_manual.pdf`, section "Changes in v1.4") is updated accordingly.

1 Summary

Change	numbers change?	size	old behaviour
GRS99 pion PDFs fixed (**`grv_version = 99`**)	yes	$F_2^{(\pi N)}$ was $\approx 2\times$ too large	not reproducible (was wrong)
Default model `exp_s`, $\Lambda = 1.56$ GeV (was Pauli–Villars, 1.33 GeV)	yes (only runs relying on defaults)	old default 1.6–1.9× larger	**`form_factor = pauli_villar`**, **`lambda = 1.33`**
kbin: **exact** integration, **new default**	yes (kbin scans)	median $\leq 0.3\%$; 2–6% near curve ends; up to 35% in the 60–100 MeV bin	**`kbin_method = grid`**, `-kbin-grid`, `pioncloud_set_kbin_method(0)`
Odd numbers of integration steps	yes (odd n only)	was $\sim 0.4\%$ wrong	not reproducible (was wrong)
Δ channels + **`dissociation = neutral`** rejected	no	was a silent $F_2 = 0$	—
$\rho\Delta$ channel selectable	no (new)	4–13% of $\pi\Delta$	—
`charge_states = total`	no (new, opt-in)		default `single`
`q2_fixed`	no (new, opt-in)		default $0 = Q^2(x)$
`pion_strange`	no (new, opt-in)		default `false`
Pluggable models (`Models.hh`)	no	bit-identical	—
Command-line options that did not work	no	9 options repaired	—
Test suite runs completely	no	80 → 216 tests actually run	—
Validation test vs. published figures	no	28 checks	—
Code documentation, `CONTRIBUTING.md`	no		—
$\pi\Delta$ normalisation note	no (documented)	model difference, factor 2.2	—

Rule followed for every change: a change either reproduces all 23 validation runs (inputs of the TDIS figures) *bit for bit*, or it changes numbers on purpose, with a reference, a measured effect and a test that fails on the old code.

2 Decisions taken (C. Ayerbe, 12 Nov 2026)

1. The new exact kbin integration becomes the **default**; it is marked as a new feature everywhere, and the original Fortran algorithm stays available behind an explicit switch (Sec. 4).
2. The $\pi\Delta$ normalisation is **kept** as in the Fortran; the difference with McKenney *et al.* is documented in a separate note (Sec. 7).
3. The documentation update of the manual (figures, validation) is merged into `main`; all code changes are on the branch `v1.4-improvements`.
4. LHAPDF backend and Python bindings: on hold.

3 Bug fixes that change numbers

3.1 GRS99 pion PDFs (`grv_version = 99`)

What. `include/GRV.hh`: the 1999 set (Glück, Reya, Schienbein, EPJ C10 (1999) 313, hep-ph/9903288, Appendix A) is implemented as published: total valence, light sea, strange sea, gluon. The per-flavour interface used by the calculation has the same meaning as for GRV92.

Why. The v1.3 routine (translated from `f2_pi_sub.f90`, where its call was commented out and never used) had three errors: (1) GRS99 define $v^\pi = u_v^\pi + \bar{d}_v^\pi$ (both valence quarks, $\int v\,dx = 2$), but it was used as one flavour in $F_2^\pi = \frac{5}{9}(xV + 2xS)$, doubling the valence part; (2) the coefficients used as "light sea" are those of the *strange* sea (their Eqs. A.7–A.8); (3) $N = 1.212 + 0.498s + 0.009s^2$ had the wrong sign.

Evidence. From the code's own formulas: GRV92 $\int V dx = 0.99$, v1.3 GRV99 $\int V dx = 1.99$. After the fix GRS99 satisfies its sum rules: $\int v\,dx = 2.00$ and the momentum sum $\int x(v + 4\bar{q} + 2\bar{s} + g)dx = 1.000 \pm 0.005$ for $Q^2 = 1$–100 GeV^2 (with a per-flavour v it would be ≈ 1.4). New unit tests check this; they fail on the v1.3 code.

Effect. Only with **`grv_version = 99`**: $F_2^{(\pi N)}$ was +107% too large on the Fig. 37 configuration; now GRS99 differs from GRV92 by +5 to +17% (a genuine PDF-set difference). Earlier GRV99 results (e.g. the 2020 `GRV91vsGRV99` comparisons) must be redone.

3.2 Default form factor and cut-off

What. Defaults are now **`form_factor = exp_s`**, **`lambda = 1.56`**, **`lambda_delta = 1.39`** (HSS central values).

Why. v1.3 defaulted to Pauli–Villars with $\Lambda = 1.33$ GeV. In McKenney *et al.* (PRD 93, 054011, Table I) 1.33 GeV is the fit of the s-dependent *exponential*; the Pauli–Villars fit is 0.27 GeV. The pairing matches no fit; **`lambda_delta = 1.39`** was already the HSS `exp_s` value; the C interface already defaulted to `exp_s`/1.56; and HSS `exp_s` is the model of the TDIS proposal figures. A cut-off belongs to its regulator.

Effect. Only runs that did not set **`form_factor`**/**`lambda`**: the old default gave $F_2^{(\pi N)}$ 1.6–1.9× the HSS model. A warning is now printed when Λ is far outside the published range of the chosen regulator.

3.3 Odd numbers of integration steps

Simpson's rule needs an even number of intervals; v1.3 accepted odd values silently (−0.4% for **`nk = 201`**). Now odd n uses a Simpson 3/8 tail (exact for cubics). Even n: unchanged, bit for bit.

4 The new kbin integration (default)

What. With $u = k_T^2$, $A = 2M(1-y)$, $c = y(2-y)M^2$, the kinematics of `f2_pi_sub.f90` give $|\vec{k}|^2 = u + (u+c)^2/A^2$ and $\cos\phi_k = (u+c)/(A|\vec{k}|)$. The $|\vec{k}|$ window and both angular cuts become

quadratic inequalities in u; at fixed y the allowed region is at most two intervals, and nothing contributes above $y_{\max}(k_{\max})$. The region is integrated with Gauss–Legendre (`include/Quadrature.hh`, `Calculator::kbinF2Exact`).

Why. The original algorithm puts k_T on a uniform grid over $[0, 10]$ GeV and rejects points outside the cuts: the integrand becomes a step function, $> 96\%$ of the points are wasted for $|\vec{k}| < 0.4$ GeV, and the error is concentrated near the kinematic end point.

Evidence (deviation from the exact method at high resolution, points within $\times 1000$ of the maximum):

configuration	grid 100×5000 (Fortran)	grid $400 \times 200\,000$ (~ 15 s)	exact (0.3 s)
Fig. 5, $\lvert\vec{k}\rvert = 60$–100 MeV	up to 100%	7.8%	0.0006%
Fig. 6, πN	13.5%	0.78%	0.002%
Fig. 37, πN	5.0%	0.25%	0.002%

Against the published TDIS curves, the maximum deviation halves (Fig. 6: 6–7% $\to$ 3.5%; Fig. 37: $\sim 2.5\% \to$ 1–1.6%). Tests: Gauss–Legendre exactness, the analytic region against a brute-force cut test on 9 600 random points, exact vs. converged grid ($< 0.5\%$), and the Fortran reference values with both methods.

Effect of the new default (17 kbin validation runs, exact vs. grid at the Fortran settings): median $\leq 0.3\%$ per curve; typically 2–6% near the end of each curve; up to 35% for the narrow 60–100 MeV bin, where the grid is not converged. x and theta scans are not affected.

To use the ORIGINAL Fortran algorithm (reproduces v1.3 bit for bit):

input file	**`kbin_method = grid`**
command line	`-kbin-grid` or `-kbin-method grid`
C / Python (ctypes)	`pioncloud_set_kbin_method(0);`
C++	`params.kbin_method = PionCloud::KBIN_GRID;`

The method in use is printed by every kbin run and written to `<base>_params.txt` and the header of `<base>_xscan.dat`. **`ny`**, **`nkt`** and **`precision`** act only on the grid method; the exact method uses **`gl_nodes`** (16) and **`gl_ypanels`** (32).

5 Other fixes and new features (no change of default numbers)

Test suite. With CTEQ6 the tests ran in a directory without the table; the Fortran `STOP` ended the suite with exit status 0 after ~ 80 tests and `ctest` reported "Passed". Now the tests run where the tables are, `ctest` requires the summary line "0 failed", and the Calculator throws a C++ exception (instead of letting the Fortran stop the host program) if the table is missing. 216 unit tests run and pass.

Validation test. `PionCloudValidation`: 28 checks against the curves of PR12-15-006 Figs. 5, 6, 37, 46, 47 (kbin figures with both methods). Shifting Λ by -1σ makes most checks fail.

Command line. `-grv`, `-nucleon`, `-precision`, `-inclusive`, `-verbose`, `-xmin-th`, `-xmax-th`, `-kmin-th`, `-kmax-th` were listed by `-help` but exited with "Unknown option"; they work now. New: `-quiet`, `-strange`, `-kbin-grid`, `-kbin-method`.

Δ + neutral. **`pi_delta`**/**`rho_delta`** with **`dissociation = neutral`** returned $F_2 = 0$ silently (zero coupling in the Fortran); now rejected with an explanation.

$\rho\Delta$. **`contribution = rho_delta`**: the Fortran $\rho\Delta$ function was ported but unreachable. Not validated against a published tagged curve.

`charge_states = total`. Sum over charge states for πN, ρN ($\times 3$ neutral, $\times \frac{3}{2}$ charge exchange): the "fully integrated" curves of the figures (previously scaled by hand).

`q2_fixed`. Fixed Q^2 at every x instead of $Q^2(x)$ at fixed electron angle.

`pion_strange`. Adds $\frac{2}{9}x\bar{s}$ to F_2^π (off by default, as in the Fortran).

Pluggable models. `include/Models.hh`: interfaces `SplittingModel` (f(y, k_T)) and `PionSFModel` ($F_2^M(x, Q^2)$). A new splitting function (chiral EFT, kaon cloud, ...) or meson PDF is one class, injected into the Calculator or registered by name (**`splitting_model`**, **`pion_sf_model`**), without editing the integration code. Built-ins are bit-identical to v1.3.

Output. The x-scan file is `<base>_xscan.dat` (v1.3 wrote `<base>_pi-N (flag=0)_kbin.dat`), columns are separated, labels correct; the parameter file records all v1.4 options.

Documentation. Physics block comments with sources in the headers (normalisation, conventions, couplings, isospin, "how to add a channel", "how to modify"); `CONTRIBUTING.md`; `CHANGELOG.md`; manual updated; the CTEQ6 `xfx` comment corrected (f, not xf).

6 Verification of PionCloud v1.4

1. **Published figures**: PR12-15-006 Figs. 5, 6, 37, 46, 47 reproduced to $\approx 1\%$ (median), curves extracted exactly from the vector graphics (automatic test).
2. **Original Fortran**: `3Var_theta.f`, `3Var_x.f` agree to 0.07–0.16% on the same grid once the Fortran end-point Simpson weight is aligned (which also explains why the published curves lie $\sim 1\%$ above PionCloud).
3. **Independent publication**: $\langle n \rangle_{\pi N} = 0.224$ / 0.310 vs. 0.23 / 0.31 in McKenney *et al.* Table I.
4. **Sum rules** for the pion PDFs.
5. **Bit-for-bit regression**: every change that should not move numbers reproduces all 23 validation runs exactly.

7 $\pi\Delta$ normalisation

$\langle n \rangle_{\pi\Delta} = 0.027$ vs. 0.06 in McKenney *et al.* for the same regulator. Implementing their Eq. (5) with the same regulator gives 0.058, so their number is reproduced; the factor 2.2 is $\times 1.5$ isospin bookkeeping (interpretation of the Fortran factor $\frac{2}{3}$) times $\times 1.43$ functional form (Sullivan Δ-pole form of the Fortran/HSS vs. chiral-EFT on-shell form); couplings agree to 4%. It is a model difference; PionCloud keeps the Fortran model. Full note: `docs/notes/piDelta_normalisation.pdf` (reproducer `docs/notes/piDelta_check.cc`).

8 Migration: reproducing v1.3 results

```
# add to a v1.4 input file to reproduce a v1.3 run
kbin_method = grid          # original Fortran kbin algorithm
form_factor = pauli_villar  # ONLY if the v1.3 run relied on the default model
lambda      = 1.33          #   (physically inconsistent pairing, see Sec. 3.2)
```

Not reproducible on purpose: **`grv_version = 99`** and odd step counts (both were wrong). Everything else gives identical numbers.

9 Open items

- Charge-state meaning of the Fortran $\pi\Delta$ factor $\frac{2}{3}$.

- $F_2^{p,n}$ reference curves of the proposal not reproduced by any CTEQ6 set (3–7% offset; affects only those reference curves and ratios).
- Sum over channels in one run; $|\vec{k}|$-scan and t-scan modes.
- On hold: LHAPDF backend, Python bindings.
- CTEQ6 thread safety (Fortran `COMMON` blocks).

Commits (branch `v1.4-improvements`)

`92dbe76`	GRS99 fix and sum-rule tests
`fe23d7f`	defaults, Δ/neutral, odd n, CLI options, ctest false pass
`26ebc7f`	pluggable models, physics documentation in the code
`03c1fb0`	exact kbin integration, **`q2_fixed`**, **`charge_states`**, validation test
`a55957a`	manual and README for v1.4
(this commit)	exact integration as default + switches, $\pi\Delta$ note, this document